



Question 1) Unitary operators are those for which $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{1}$.

- Prove that if $\hat{U}$ is unitary, that its eigenvalues have absolute value 1.
- Writing $\hat{U} = e^{i\hat{H}}$ for some operator $\hat{H}$, show that $\hat{U}$ is unitary when $\hat{H}$ is Hermitian. (Expand $e^{i\hat{H}}$).
- For any two orthonormal bases $\{|a_i\rangle\}$ and $\{|b_i\rangle\}$, show that there exists a unitary operator $\hat{U}$ such that $\hat{U}|a_i\rangle = |b_i\rangle$ for all i . What is it?
- Show that as far as all the matrix elements of an operator are concerned, a **unitary** transformation of bases $|a_i\rangle \mapsto |b_i\rangle = \hat{U}|a_i\rangle$ is equivalent to a **similarity transformation** of the operator, i.e., if $\hat{A} \mapsto \hat{A}' = \hat{U}\hat{A}\hat{U}^\dagger$.
- Show that a unitary transformation of bases preserves:
 - inner products, i.e., for any vectors $|\psi\rangle$ and $|\phi\rangle$, $\langle\psi|\phi\rangle = \langle\hat{U}\psi|\hat{U}\phi\rangle$.
 - norms, i.e., for any vector $|\psi\rangle$, $\| |\psi\rangle \| = \| \hat{U} |\psi\rangle \|$.
 - commutators, i.e., for any two operators $\hat{A}$ and $\hat{B}$, $\langle[\hat{A}, \hat{B}]\rangle_\psi = \langle[\hat{U}\hat{A}\hat{U}^\dagger, \hat{U}\hat{B}\hat{U}^\dagger]\rangle_{\hat{U}\psi}$.
 - eigenvalues, i.e., for any operator $\hat{A}$ with eigenvalues λ_i , the eigenvalues of $\hat{U}\hat{A}\hat{U}^\dagger$ are also λ_i with the transformed eigenvector $\hat{U}|\psi_i\rangle$ where $\hat{A}|\psi_i\rangle = \lambda_i|\psi_i\rangle$.
 - traces, i.e., for any operator $\hat{A}$, $\text{Tr}(\hat{A}) = \text{Tr}(\hat{U}\hat{A}\hat{U}^\dagger)$.

Which of these hold for non-unitary similarity transformations? ($\hat{A} \mapsto \hat{A}' = \hat{S}\hat{A}\hat{S}^{-1}$ with $\hat{S}$ not necessarily unitary but invertible.)



Read S&N Sec. 1.7.2 - 1.7.5.



Question 2) Consider a more general operator $\hat{V} = e^{i\hat{G}}$ for some $\hat{G} = \hat{H} + i\hat{\Gamma}$, where both $\hat{H}$ and $\hat{\Gamma}$ are Hermitian. Assume that $[\hat{H}, \hat{\Gamma}] = 0$, so that they share a common eigenbasis.

- Under what condition is $\hat{V}$ unitary?
- From now on, consider the special case when $\hat{\Gamma} = \Gamma\hat{1}$ for some real constant Γ . Interpreting $\hat{H} + i\hat{\Gamma}$ as a complexified momentum operator (i.e., $\hat{p} \mapsto \hat{\mathcal{P}} = \hat{p} + i\hat{\Gamma}$), how does the translation operator

$$\hat{\mathcal{J}}(\Delta x) \rightarrow \lim_{N \rightarrow \infty} \left(\hat{1} - i \frac{\hat{\mathcal{P}} \Delta x}{N\hbar} \right)^N \quad (1)$$

act on a state $|\alpha\rangle$? What happens to the norm of the state under translations when $\hat{\Gamma} \neq 0$?

- Find the general expression for the “complexified” momentum operator $\hat{p} + i\hat{\Gamma}$ on a state α , generalizing S&N Eq. 1.249 and S&N Eq. 1.252. That is, find the generalization of

$$\langle x' | \hat{p}^n | \alpha \rangle = (-i\hbar)^n \frac{\partial^n}{\partial x'^n} \langle x' | \alpha \rangle = (-i\hbar)^n \frac{\partial^n}{\partial x'^n} \psi_\alpha(x') \quad \text{for } \hat{p} \mapsto \hat{\mathcal{P}} \quad (2)$$

- What is the expectation value of the momentum $\langle \hat{\mathcal{P}} \rangle_\alpha$ and kinetic energy of our state $\langle \hat{T} \rangle_\alpha = \langle \frac{\hat{\mathcal{P}}^2}{2m} \rangle_\alpha$ in terms of $\langle \hat{p} \rangle_\alpha$ and $\langle \hat{p}^2 \rangle_\alpha$? Can you reflect on what this would mean physically? (Hint: we will see later that for a free-particle the time evolution of a state is given by $|\alpha(t)\rangle = e^{-i\hat{T}t/\hbar} |\alpha\rangle$.)

1 Question 1

Unitary operators are like “rotations” of Hilbert space. They move kets around without changing any of the geometry (no stretching or twisting). See S&N Section 1.5 for the change-of-basis discussion.

a) Consider the eigenbasis of our unitary operator for which $\hat{U}|u_i\rangle = u_i|u_i\rangle$. Taking the Hermitian conjugate gives $\langle u_i|\hat{U}^\dagger = u_i^*\langle u_i|$, and so

$$\langle u_i|u_i\rangle = \langle u_i|\hat{U}^\dagger\hat{U}|u_i\rangle = u_i^*u_i\langle u_i|u_i\rangle = |u_i|^2\langle u_i|u_i\rangle. \quad (3)$$

Since $\langle u_i|u_i\rangle > 0$, $|u_i|^2 = 1$. Given that any complex number can be written as $u_i = \rho_i e^{i\theta_i}$ with $\rho_i \geq 0$ and real θ_i , the condition $|u_i|^2 = 1$ gives $\rho_i = 1$. Hence $u_i = e^{i\theta_i}$ for some real θ_i .

i

We will see/have seen the (unitary) time-evolution operator $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$ in S&N Section 2.1. Its eigenkets are the energy eigenkets $|E_n\rangle$ and its eigenvalues are the phases $e^{-iE_nt/\hbar}$. Because time evolution is unitary in QM, its eigenvalues had to be phases.

i

There is one more nice property that we did not discuss that is relevant for unitary operators: they are always **normal**. A normal operator is one that commutes with its Hermitian conjugate:

$$[\hat{N}, \hat{N}^\dagger] = 0. \quad (4)$$

The above property is obviously true for a unitary $\hat{U}$. Normal operators have the nice property that they can be diagonalized in an orthonormal basis, so we are okay to assume, for example, that the energy basis, for example, is orthonormal: it is the eigenbasis of the Hamiltonian and also of the time evolution operator, both of which are normal operators. You can prove the orthonormality property by the same argument we used in HW 1, where we had $(\lambda - \lambda')\langle u'|u\rangle = 0$.

b) If we write $\hat{U} = e^{i\hat{H}}$ and Taylor expand the exponential,

$$\begin{aligned} \hat{U}^\dagger\hat{U} &= \left(\hat{1} - i\hat{H}^\dagger - \frac{1}{2}\hat{H}^{\dagger 2} + \dots\right) \left(\hat{1} + i\hat{H} - \frac{1}{2}\hat{H}^2 + \dots\right) \\ &= \hat{1} + i(\hat{H} - \hat{H}^\dagger) - \left(\frac{\hat{H}^2 + \hat{H}^{\dagger 2}}{2} - \hat{H}^\dagger\hat{H}\right) + \dots \end{aligned} \quad (5)$$

You can check that this is equal to $\hat{1}$ order by order in $\hat{H}$ when $\hat{H} = \hat{H}^\dagger$. Conversely, if $\hat{H}$ is Hermitian then $\hat{U}^\dagger = e^{-i\hat{H}}$, which commutes with $e^{i\hat{H}}$ (they are both functions of the same operator), and $e^{-i\hat{H}}e^{i\hat{H}} = e^0 = \hat{1}$ exactly, to all orders. So a Hermitian $\hat{H}$ is sufficient for $\hat{U}$ to be unitary.

i

This is quite general for **generators** of unitary operators or **unitary groups**. Consider a one-parameter family of transformation like $\hat{U}(\epsilon) = e^{i\epsilon\hat{H}}$ that is unitary for any real ϵ (a one-parameter unitary group). If we differentiate $\hat{U}^\dagger(\epsilon)\hat{U}(\epsilon) = \hat{1}$ at $\epsilon = 0$, we get $-i\hat{H}^\dagger + i\hat{H} = 0$, and $\hat{H}$ must be Hermitian. This is called Stone’s theorem and it tells us that every continuous symmetry in quantum mechanics (unitary groups like translations, rotations, time evolution) comes with a Hermitian generator (momentum, angular momentum, the Hamiltonian). For continuous symmetries, the statement is necessary and sufficient: every continuous unitary symmetry has a Hermitian generator, and every Hermitian operator generates a continuous unitary symmetry.



By the way, you should note that in general, the product of exponential of operators need not obey our usual rule $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}$. The more general statement is called the **Baker-Campbell-Hausdorff** formula (this I am sure will come back to haunt you at some point):

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]+\frac{1}{12}[\hat{A},[\hat{A},\hat{B}]]-\frac{1}{12}[\hat{B},[\hat{A},\hat{B}]]+\dots} \quad (6)$$

The above recovers the familiar rule $e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}$ when $[\hat{A},\hat{B}] = 0$. The general case is a lot more complicated, but we almost never need all terms because in many applications each additional power of the operators correspond to a higher order term in some expansion, and we end up truncating:

$$e^{\hat{A}}e^{\hat{B}} \approx e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]}. \quad (7)$$

Another beautiful case is when A and B commute with their commutators... for example, if $\hat{A} = \hat{x}$ and $\hat{B} = \hat{p}$, then $[\hat{x},\hat{p}] = i\hbar\mathbb{1}$, and $[\hat{x},[\hat{x},\hat{p}]] = [\hat{p},[\hat{x},\hat{p}]] = 0$. In this case, you'd have

$$e^{a\hat{x}/\hbar}e^{b\hat{p}/\hbar} = \exp\left[i\frac{a\hat{x}+b\hat{p}}{\hbar} - \frac{iab}{2\hbar}\right], \quad (8)$$

which holds exactly.

Finally, let's try the case of angular momentum. As we will see, rotation operators in quantum mechanics are (as you can already guess...), $e^{-i\theta\hat{\mathbf{n}}\cdot\hat{\mathbf{J}}/\hbar}$, where $\hat{\mathbf{n}}$ is the axis of rotation and $\hat{\mathbf{J}}$ is the angular momentum operator vector. So, a rotation around the x axis is $e^{-i\theta\hat{J}_x/\hbar}$ for example (I am setting $\hbar = 1$ for my own sanity in what follows). Take $[\hat{J}_x, \hat{J}_y] = i\hat{J}_z$, and the commutators $[\hat{J}_x, [\hat{J}_x, \hat{J}_y]] = i\hat{J}_y$ and $[\hat{J}_y, [\hat{J}_x, \hat{J}_y]] = -i\hat{J}_x$. Two consecutive rotations, one around x and one around y , amount to

$$e^{-i\hat{J}_x\theta_x}e^{-i\hat{J}_y\theta_y} = \exp\left[-i(\hat{J}_x\theta_x + \hat{J}_y\theta_y + \frac{1}{2}\theta_x\theta_y\hat{J}_z - \frac{\theta_x\theta_y}{12}(\theta_y\hat{J}_x + \theta_x\hat{J}_y) + \dots)\right], \quad (9)$$

where I used

$$[\theta_x\hat{J}_x, [\theta_x\hat{J}_x, \theta_y\hat{J}_y]] - [\theta_y\hat{J}_y, [\theta_x\hat{J}_x, \theta_y\hat{J}_y]] = \theta_x^2\theta_y\hat{J}_y - (-\theta_x\theta_y^2\hat{J}_x) = \theta_x^2\theta_y\hat{J}_y + \theta_x\theta_y^2\hat{J}_x. \quad (10)$$

The price we pay to use a single exponential is an awful exponent to be evaluated. Because rotations form a group, it must be the case that the product of two rotation operators is itself a rotation operator, i.e.,

$$e^{-i\hat{J}_x\theta_x}e^{-i\hat{J}_y\theta_y} = e^{-i\hat{\mathbf{n}}\cdot\hat{\mathbf{J}}\theta}, \quad (11)$$

for some axis $\hat{\mathbf{n}}$ and angle θ . It is just hard to see what $\hat{\mathbf{n}}$ and θ are in terms of θ_x and θ_y , but it exists. None of this is a big deal if the angles involved are small, though:

$$e^{-i\hat{J}_x\theta_x}e^{-i\hat{J}_y\theta_y} \approx \exp\left[-i(\hat{J}_x\theta_x + \hat{J}_y\theta_y) + \mathcal{O}(\theta_x\theta_y)\right], \quad (12)$$

which is just the fact that small rotations in 3D approximately commute.

c) Looking at S&N Section 1.5, the operator that sends each $|a_i\rangle$ to $|b_i\rangle$ is

$$\hat{U} = \sum_i |b_i\rangle \langle a_i|. \quad (13)$$

It had to be so: $\hat{U}|a_j\rangle = \sum_i |b_i\rangle \langle a_i|a_j\rangle = \sum_i |b_i\rangle \delta_{ij} = |b_j\rangle$ by orthonormality of the $\{|a_i\rangle\}$. It

is unitary by orthonormality and completeness of both bases:

$$\begin{aligned}\hat{U}^\dagger \hat{U} &= \sum_{i,j} |a_i\rangle \langle b_i| b_j\rangle \langle a_j| = \sum_{i,j} |a_i\rangle \delta_{ij} \langle a_j| = \sum_i |a_i\rangle \langle a_i| = \hat{\mathbb{1}}, \\ \hat{U} \hat{U}^\dagger &= \sum_{i,j} |b_i\rangle \langle a_i| a_j\rangle \langle b_j| = \sum_i |b_i\rangle \langle b_i| = \hat{\mathbb{1}}.\end{aligned}\tag{14}$$

d) Sandwiching $\hat{U}^\dagger \hat{U} = \hat{\mathbb{1}}$ on both sides of $\hat{A}$,

$$\langle a_i | \hat{A} | a_j \rangle = \langle a_i | \hat{U}^\dagger (\hat{U} \hat{A} \hat{U}^\dagger) \hat{U} | a_j \rangle = \langle b_i | \hat{A}' | b_j \rangle, \quad \hat{A}' = \hat{U} \hat{A} \hat{U}^\dagger. \tag{15}$$

In other words: the matrix of $\hat{A}$ in the old basis is the same as the matrix of $\hat{A}' = \hat{U} \hat{A} \hat{U}^\dagger$ in the new basis.



There is an important distinction between how we act $\hat{U}$ on states versus operators that is reminiscent of active versus passive rotations in classical mechanics. You will probably never remember the sign of the rotation angle until you need it, but the good news is that it is easy to see which is the right one for your application by just trying it out.

A passive transformation amounts to “rotating the kets”, we can identify the transformed operator as $\hat{A} \mapsto \hat{U}^\dagger \hat{A} \hat{U}$. An active transformation “rotates the operator”, in which case we identify $\hat{A} \mapsto \hat{U} \hat{A} \hat{U}^\dagger$ in Eq. (15). Note the location of the daggers.

The key is: if you transform both the states, $|\psi\rangle \mapsto \hat{U} |\psi\rangle$, and the operators, $\hat{A} \mapsto \hat{U} \hat{A} \hat{U}^\dagger$, then all matrix elements stay the same, and no physics changes. For rotations, this statement is obvious: rotate the vector and the coordinate system by the same angle, and the components of the vector in the new system remain the same.

This is the key idea we have in mind when we end up saying that $\hat{U}$ is a symmetry (in the future, S&N Section 4.1).

e)

- i) $\langle \hat{U} \psi | \hat{U} \phi \rangle = \langle \psi | \hat{U}^\dagger \hat{U} | \phi \rangle = \langle \psi | \phi \rangle$.
- ii) $\| \hat{U} | \psi \rangle \|^2 = \langle \psi | \hat{U}^\dagger \hat{U} | \psi \rangle = \| | \psi \rangle \|^2$ (this is what we mean by “unitary transformation conserves probabilities”).
- iii) $[\hat{U} \hat{A} \hat{U}^\dagger, \hat{U} \hat{B} \hat{U}^\dagger] = \hat{U} \hat{A} \hat{U}^\dagger \hat{U} \hat{B} \hat{U}^\dagger - \hat{U} \hat{B} \hat{U}^\dagger \hat{U} \hat{A} \hat{U}^\dagger = \hat{U} (\hat{A} \hat{B} - \hat{B} \hat{A}) \hat{U}^\dagger = \hat{U} [\hat{A}, \hat{B}] \hat{U}^\dagger$. If we define $\hat{C} = [\hat{A}, \hat{B}]$, then the statement above is what we expect: $\hat{C} \mapsto \hat{C}' = \hat{U} \hat{C} \hat{U}^\dagger$. Therefore, from Eq. (15), we already know that expectation values are conserved:

$$\langle \hat{C} \rangle_\psi = \langle \psi | [\hat{A}, \hat{B}] | \psi \rangle = \langle \hat{U} \psi | [\hat{A}', \hat{B}'] | \hat{U} \psi \rangle = \langle \hat{C}' \rangle_{\hat{U} \psi}. \tag{16}$$



A trivial example is $[\hat{x}, \hat{p}] = i\hbar \hat{\mathbb{1}}$, which actually remains the same under any unitary transformation because $\hat{U} \hat{\mathbb{1}} \hat{U}^\dagger = \hat{\mathbb{1}}$. In this case not just the expectation values, but the commutation relation itself remains unchanged.

- iv) Start with $\hat{A} | a \rangle = \lambda | a \rangle$. Multiply both sides by $\hat{U}$ and add a $\hat{U}^\dagger \hat{U}$ in the right place to get

$$(\hat{U} \hat{A} \hat{U}^\dagger) (\hat{U} | a \rangle) = \hat{U} \hat{A} | a \rangle = \lambda (\hat{U} | a \rangle), \tag{17}$$

so $\hat{U}|a\rangle$ is an eigenket of $\hat{A}'$ with the same eigenvalue λ . Unitary operators rotate the eigenkets but leave the eigenvalues the same. This had to be the case: the spectrum (i.e., all the possible outcomes of a measurement) should not depend on the basis you use to describe it.

$$v) \text{Tr}(\hat{U}\hat{A}\hat{U}^\dagger) = \text{Tr}(\hat{U}^\dagger\hat{U}\hat{A}) = \text{Tr}(\hat{A})$$



Another way to see it is that the Trace of an operator is the sum of its eigenvalues. But we just showed the eigenvalues are invariant, so the trace must also be invariant. The same reasoning actually applies to the determinant (the product of all eigenvalues), so we could also have proven that $\det(\hat{U}\hat{A}\hat{U}^\dagger) = \det(\hat{U})\det(\hat{A})\det(\hat{U}^\dagger) = \det(\hat{U})\det(\hat{U}^\dagger)\det(\hat{A}) = \det(\hat{U}\hat{U}^\dagger)\det(\hat{A}) = \det(\hat{A})$.

Note: the first three items actually need $\hat{U}$ to be unitary (because $(|U\psi\rangle)^\dagger = \langle\psi|U^\dagger \neq \langle\psi|U^{-1}$ in general). It is true, however, that the structure of the commutation relations is preserved: $[\hat{A}', \hat{B}'] = \hat{S}[\hat{A}, \hat{B}]\hat{S}^{-1}$, but this is technically a different operator.

2 Question 2

We now break away from unitary operators to see what would happen. The main message is that a non-unitary transformation does not preserve norms, so “probability is no longer conserved”. Sometimes this may be intentional, like when you need to describe systems with decaying states, particles that are absorbed, or injected into a system.

Throughout, we will use the fact that $[\hat{H}, \hat{\Gamma}] = 0$ so that we can split the exponential, $\hat{V} = e^{i\hat{H}}e^{-\hat{\Gamma}}$.

a) Since $\hat{H}$ and $\hat{\Gamma}$ are Hermitian, $\hat{G}^\dagger = \hat{H} - i\hat{\Gamma}$, and by the same argument as in 1b), $\hat{V}^\dagger = e^{-i\hat{G}^\dagger} = e^{-i\hat{H} + \hat{\Gamma}}$. Because $\hat{H}$ and $\hat{\Gamma}$ commute, we can readily write $\hat{V}^\dagger\hat{V} = e^{-2\hat{\Gamma}}$. In the eigenbasis of $\hat{\Gamma}$, we have

$$\hat{V}^\dagger\hat{V} = e^{-2\hat{\Gamma}} = \sum_i e^{-2\gamma_i} |\gamma_i\rangle \langle\gamma_i|, \quad (18)$$

where $\hat{\Gamma}|\gamma_i\rangle = \gamma_i|\gamma_i\rangle$ and γ_i are real. So for $\hat{V}$ to be unitary, we need $\hat{V}^\dagger\hat{V} = \mathbb{1}$, which requires $e^{-2\gamma_i} = 1$ for all i , or $\gamma_i = 0$ for all i . An operator whose eigenvalues are all null is $\hat{\Gamma} = 0$.



This form we used is called the polar decomposition of $\hat{V}$: it is a unitary $e^{i\hat{H}}$ times a positive Hermitian “stretch” factor $e^{-\hat{\Gamma}}$. Every operator can be written this way, $\hat{V} = \hat{U}\hat{R}$ with $\hat{R} = \sqrt{\hat{V}^\dagger\hat{V}}$, in analogy with $z = e^{i\theta}|z|$ for complex numbers.

b) With $\hat{\mathcal{P}} = \hat{p} + i\Gamma$, the limit of infinitely many applications of the translation operator gives an exponential exactly as in S&N Section 1.6:

$$\hat{\mathcal{J}}(\Delta x) = \lim_{N \rightarrow \infty} \left(\mathbb{1} - i \frac{\hat{\mathcal{P}}\Delta x}{N\hbar} \right)^N = e^{-i\hat{\mathcal{P}}\Delta x/\hbar} = e^{-i\hat{p}\Delta x/\hbar} e^{\Gamma\Delta x/\hbar} = e^{\Gamma\Delta x/\hbar} \hat{\mathcal{J}}(\Delta x), \quad (19)$$

where $\hat{\mathcal{T}}(\Delta x) = e^{-i\hat{p}\Delta x/\hbar}$ is the usual unitary translation operator and I could split the exponential because $\Gamma\hat{1}$ commutes with everything. Acting it on a state,

$$\hat{\mathcal{J}}(\Delta x)|\alpha\rangle = e^{\Gamma\Delta x/\hbar}\hat{\mathcal{T}}(\Delta x)|\alpha\rangle, \quad \langle x'|\hat{\mathcal{J}}(\Delta x)|\alpha\rangle = e^{\Gamma\Delta x/\hbar}\psi_\alpha(x' - \Delta x). \quad (20)$$

In addition to shifting the argument of the wavefunction by Δx as before, we multiply it by a real factor $e^{\Gamma\Delta x/\hbar}$. The norm is not preserved:

$$\langle\alpha|\hat{\mathcal{J}}^\dagger(\Delta x)\hat{\mathcal{J}}(\Delta x)|\alpha\rangle = e^{2\Gamma\Delta x/\hbar}\langle\alpha|\hat{\mathcal{T}}^\dagger\hat{\mathcal{T}}|\alpha\rangle = e^{2\Gamma\Delta x/\hbar}\langle\alpha|\alpha\rangle, \quad (21)$$

For $\Gamma > 0$, translating to the right amplifies the state exponentially and translating to the left damps it.



It is interesting that $\hat{\mathcal{J}}$ still satisfies the properties of a group: $\hat{\mathcal{J}}(a)\hat{\mathcal{J}}(b) = \hat{\mathcal{J}}(a+b)$ and $\hat{\mathcal{J}}(\Delta x)^{-1} = \hat{\mathcal{J}}(-\Delta x)$. The only property that we lose is $\hat{\mathcal{J}}^{-1} = \hat{\mathcal{J}}^\dagger$. This means these non-unitary translations are still a group, but just not a unitary one.

c) Since Γ is just a number, $\langle x'|\hat{\mathcal{P}}|\alpha\rangle = \langle x'|\hat{p}|\alpha\rangle + i\Gamma\langle x'|\alpha\rangle$, i.e.,

$$\langle x'|\hat{\mathcal{P}}|\alpha\rangle = \left(-i\hbar\frac{\partial}{\partial x'} + i\Gamma\right)\psi_\alpha(x') = -i\hbar\left(\frac{\partial}{\partial x'} - \frac{\Gamma}{\hbar}\right)\psi_\alpha(x'), \quad (22)$$

which generalizes S&N Eq. (1.249). For powers, the two terms in the parentheses commute (one is a derivative, the other a constant), so we can just iterate (recalling with great effort the binomial theorem...):

$$\langle x'|\hat{\mathcal{P}}^n|\alpha\rangle = (-i\hbar)^n\left(\frac{\partial}{\partial x'} - \frac{\Gamma}{\hbar}\right)^n\psi_\alpha(x') = (-i\hbar)^n\sum_{k=0}^n\binom{n}{k}\left(-\frac{\Gamma}{\hbar}\right)^{n-k}\frac{\partial^k\psi_\alpha(x')}{\partial x'^k}, \quad (23)$$

generalizing S&N Eq. (1.252).



There is a much nicer way of writing this. Check that

$$\left(\frac{\partial}{\partial x'} - \frac{\Gamma}{\hbar}\right)\psi(x') = e^{\Gamma x'/\hbar}\frac{\partial}{\partial x'}\left[e^{-\Gamma x'/\hbar}\psi(x')\right] \quad (24)$$

by explicitly differentiating the right-hand side. Then, iterating, $(\partial_{x'} - \Gamma/\hbar)^n\psi = e^{\Gamma x'/\hbar}\partial_{x'}^n[e^{-\Gamma x'/\hbar}\psi]$. In other words, in terms of operators, the complexified momentum operator can be written as a **similarity transformation of the ordinary momentum operator**:

$$\hat{\mathcal{P}} = e^{\Gamma\hat{x}/\hbar}\hat{p}e^{-\Gamma\hat{x}/\hbar}. \quad (25)$$

Note this is a non-unitary similarity transformation, but the operator $\hat{S} = e^{\Gamma\hat{x}/\hbar}$ still invertible. Any Hamiltonian built out of $\hat{\mathcal{P}}$ and $\hat{x}$ is “similar” to the corresponding Hermitian one,

$$\hat{H} = \frac{\hat{\mathcal{P}}^2}{2m} + V(\hat{x}) = \hat{S}\left(\frac{\hat{p}^2}{2m} + V(\hat{x})\right)\hat{S}^{-1}. \quad (26)$$

Note that the similarity transformation preserves the structure of the Hamiltonian, but not expectation values or norms of states.

d) It is easy to see that

$$\langle \hat{\mathcal{P}} \rangle_\alpha = \langle \hat{p} \rangle_\alpha + i\Gamma, \quad (27)$$

and expanding $\hat{\mathcal{P}}^2 = (\hat{p} + i\Gamma)^2 = \hat{p}^2 + 2i\Gamma\hat{p} - \Gamma^2$ (recall Γ is a number),

$$\langle \hat{T} \rangle_\alpha = \frac{\langle \hat{\mathcal{P}}^2 \rangle_\alpha}{2m} = \frac{\langle \hat{p}^2 \rangle_\alpha}{2m} - \frac{\Gamma^2}{2m} + i \frac{\Gamma \langle \hat{p} \rangle_\alpha}{m}. \quad (28)$$

The expectation values are both complex (so not true observables).

For a free particle, $\hat{T}$ generates time evolution, $|\alpha(t)\rangle = e^{-i\hat{T}t/\hbar} |\alpha\rangle$, then the norm of the state obeys

$$\langle \alpha(t) | \alpha(t) \rangle = \langle \alpha | e^{i\hat{T}^\dagger t/\hbar} e^{-i\hat{T}t/\hbar} | \alpha \rangle = \langle \alpha | e^{2\Gamma\hat{p}t/(\hbar m)} | \alpha \rangle. \quad (29)$$

Again, probability is not conserved.

There is one final connection I wanted to make. Take a momentum eigenstate (plane wave) for $|\alpha\rangle = |p\rangle$, such that $\hat{p}|p\rangle = p|p\rangle$ and evolve it with $\hat{T}$:

$$\langle p(t) | p(t) \rangle = \exp \left[\frac{2\Gamma p t}{\hbar m} \right]. \quad (30)$$

This means for $\Gamma > 0$, a plane wave moving to the right ($p > 0$) grows in probability, and a plane wave moving to the left ($p < 0$) has a decreasing probability. This is exactly what we saw in part c): in a time t our particle would translate $\Delta x = (p/m)t$, and Eq. (21) multiplies its norm by $e^{2\Gamma\Delta x/\hbar} \simeq 1 + (2\Gamma/\hbar)(p/m)t$. Everything agrees.



There are many examples of complex momenta and energies. For example, plane waves with complex momentum, $e^{i(\hbar k + i\Gamma)x/\hbar} = e^{i\hbar k x/\hbar} e^{-\Gamma x/\hbar}$ are attenuated waves being absorbed by a medium. This is similar to evanescent waves (wavefunctions inside a barrier when tunneling), except the latter have purely imaginary momentum (they are only decaying or growing in space, and not propagating). In scattering theory, we will see that unstable particles and resonances appear as poles of amplitudes with complex energies $E = E_R - i\Gamma_R/2$, and the imaginary part is the decay width, with $\tau = \hbar/\Gamma_R$ the lifetime. In optical and nuclear physics the absorption of a photon or neutron by the medium or nuclei is modelled with a complex “optical potential” $V - iW$, so that $\text{Im}\langle \hat{H} \rangle = -\langle W \rangle < 0$ accounts for absorption. This is equivalent to working with a complex index of refraction. We will come back to this in the future with time-dependent perturbation theory in S&N Section 5.9 and resonances in scattering (S&N Section 6.7).